06 — Mission Crane 3000 kW (budget devoured, plant reshuffles)

> **Proves** · Priority devouring the whole budget — and the anti-double-counting rule between the battery and Level 3.
>
> **Mechanics this scenario turns on**
>
> - A **Mission** row takes `H` = the heavy consumer's full rating — a crane can start at any moment, so its whole draw is a potential peak.
> - Cascade per row: `H` = what it wants · `I = min(remaining budget, H)` · `J = I × CoverageFactor` (covered) · `L = H − J` (left to the gensets). Priority: DpReserve → DpDemand → Mission → Propulsion → Hotel. Invariant `ΣJ + ΣL = ΣH` — the sea sets the swing, the battery moves the split.
> - Level 3 (DRC) damps the hotel/mission swing: `variation × 0.8`, minus whatever the battery already shaved, so the same kilowatt is never counted twice.
>
> **Panels described below** · Battery Contribution · Power Demands — the reshuffle · Baseline & IL1 · Battery Benefit & IL3
>
> **Anything not described here** behaves exactly as in `01-excel-baseline` — same plant, same hours; only what this scenario changes is worked through below.
>
> **Trust** · verified against the reference workbook. These figures are proof.
>
> **Read after** · scenario 05.

Test 01's vessel + **Mission Max = 3000** (battery 1260, Transit).

Battery Contribution

Row	H	I	J	L
Mission (D=0.5)	**3 000**	min(1260, 3000) = **1 260** (ALL of it)	**630**	**2 370**
Propulsion	573.15	min(0, …) = **0**	0	**573.15**
Hotel	76	**0**	0	**76**

Tiles: **PS = 630 · SR = 3 019.2**. The crane, third in priority, drains the budget; everyone below starves.

Power Demands — the reshuffle

Hotel' = 3 800+2 370+76 = **6 246** — MORE than one SG (3 250) can carry. Options: 1 ME+SG+AEs (SG 3 250 + AE 2 996 — expensive aux running), or **2 ME + both SGs (6 500) + 0 AE**.

The optimizer picks the second: ME = 12 036.15+6 246 = **18 282** @ 38.1 %, **AE = 0**, SG at 96.1 % (6 246/6 500). Energy ME 18 282.15×5 000 = **91 410 750**.

Baseline & IL1

Sorted: **3.1095 (2/on/0 — now FIRST!)** / 3.1292 (1/1) / **3.1780 (1/2 — baseline)** / 3.2102 / 3.2210. The combo that was the WORST row in test 01 is now the best — the uncovered reserve changed which configuration wins, not just how much fuel it burns.

- Baseline = 3.1780×5 000 = **15 890 t/yr** (AE 2 996/8 000 = 37.5 %)
- IL1 = 3.1095×5 000 = **15 547.3** · **Savings 342.7 t = $267 335** — 16× test 01's, because the candidate rows are now genuinely different machines, not ±1 idle AE.

Battery Benefit & IL3

Benefit = covered 630 monetized: **631.3 t/yr = $492 407** — the ceiling for this battery (630 = the whole 1260 × 50 %). IL3 detail: the mission covered band (630, hotel side) fully absorbs the ±500 DRC variation → Variation shows **0**, batteryShaved **500**, L3 adds **0** (anti-double-counting, rule Q4/D4).

Takeaway: the cascade runs BEFORE combination enumeration for a reason — uncovered reserve can flip the optimal plant configuration entirely.